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Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:			
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Task Description:

(1) Original Aims
The principal, long-term objective of this proposed effort is to develop a smart medical device that would be lightweight, portable, FDA-approved, under commercial development, and capable of addressing various aspects of four risks described in the Bioastronautics Roadmap [NASA/SP-2005-6113]. In particular, we seek to address under Risk 18, the risk of blunt internal trauma and internal bleeding resulting, for example, from a collision in microgravity between an astronaut and a heavy object. The device we propose would be capable of detecting internal bleeding and inducing transcutaneous hemostasis. Under Risks 18 and 28, as recent experiences in Antarctica demonstrate, malignant tumors that require some form of surgery may well appear without warning, especially in the high-radiation load of outside LEO, even when extensive pre-screening is undertaken. The device we propose would be capable of performing bloodless ablation of detectable tumors without fear of metastasis. Under Risk 4, the formation of renal calculi has been suggested as a likely result in extended space flight; indeed, a precedent exists of early mission termination due to a ureteral stone. The device we propose would be capable of detecting and sufficiently comminuting a renal calculus for it to pass without obstruction through the urinary system.

Specific Aim 1. To expand the capability of our existing smart medical device for transcutaneous acoustic hemostasis to include a fully integrated ultrasound detection, targeting, and therapy system.

Specific Aim 2. To develop the capability of the integrated image-guided therapy system in SA 1 to target and ablate specific tissue volumes such as benign and malignant tumors.

Specific Aim 3. To develop the capability of the integrated image-guided therapy system in SA 1 to target and comminute renal calculi.

(2) Key Findings

Our Specific Aims have been achieved. Smart medical systems and techniques have been developed for 1) bleeding, 2) cancer, and 3) kidney stones. Each is at the gateway of TRL/CRL 6. Army surgeons have tested the system on bleeding in animals and complementary Phase II funding from DARPA has been received to build and test in partnership with a company a specific portable hemostasis device for human use. An Initial Device Exemption (IDE) has been obtained for an ultrasound-guided HIFU system for cancer treatment and we are in the process of setting up clinical trials. A patient study has been proposed to NIH and our Human Subjects Division to test our imaging feedback and guidance systems for acoustic therapy for kidney stones. One of our feedback techniques could be used on ISS today to detect kidney stones.

In addition to moving devices to patient testing, we have these findings that above all contribute to obtaining approval for clinical use and increase the efficiency and therefore portability of the systems: • We offered a new explanation – nonlinear acoustic propagation – as to how HIFU can be used in vivo and in developing commercial systems to enhance heating. • We discovered advantages of ultrasound guidance over MRI guidance that will benefit HIFU in space where MRI is not an option.

• A novel hydrophone – a tool necessary for characterizing the output of a HIFU device - was created and characterized, and has some advantages over the existing high amplitude hydrophone.

• The gold standard hydrophone was re-calibrated to improve its accuracy at high frequencies, which are important in HIFU heating.

• A method to derate water measurements to tissue was devised and tested. • Tools, e.g., monitoring voltage to the source as a detector or localized boiling, to use the derating method in real-time were developed. • We found that tissue ablation can span a spectrum from nearly purely thermal to nearly purely mechanical based on the HIFU pulsing scheme to control boiling created by nonlinearly enhanced heating. • An automated system, which detected bleeds, distinguishing them from patent vessels, targets the bleed site, treats, and stops treatment when the bleed is no longer detected, was built and submitted as an invention disclosure and a peer-reviewed manuscript.

• An acoustic system was developed, patented, and reported to size kidney stone fragments during ureteroscopy to present the clinician or astronaut (as may be needed) from attempting to extract too large a fragment. • Following on our work to obtain an Investigation Device Exemption (IDE) last year from the U.S. FDA, a contract was secured to for a new HIFU system to begin treating patients for pancreatic cancer.

• Based on successful preliminary work benefiting from knowledge gained through NSBRI funded research, a Phase II DoD grant was secured to build an acoustic hemostasis device.

(3) Impact on the hypotheses, requirements, objectives, and aims of the proposal

The impact is that we have been able to begin work on a second generation of these systems as part of a renewal proposal and other leveraged proposals. The new systems are automated, integrated with other SMST technology, more efficient, and smaller with novel imaging feedback and are not only used for treatment but for prevention.

(4) Proposed Research Plan

This is the final year of this grant. The plans for the renewal grant have been elaborated in the funded proposal and include automated imaging to distinguish bleeding and new techniques to detect small renal calculi, to facilitate stone passage, and to prevent clinically significant stone formation. We have also applied for a NSBRI Postdoctoral Fellowship to investigate immune response to HIFU. In addition, we plan to continue playing a leadership role within the ultrasound, particularly HIFU, community to promote safe effective new therapies and guidance.

Rationale for HRP Directed Research:

	We have proposed to develop a smart medical device that would have specific applications to the risks associated with space travel, and due to the major constraints and uniqueness of this environment, this device, as presently conceived, would most probably not, in itself, be of value to Earth-based clinical medicine. However, the various technologies that are to be incorporated into this device have many important clinical applications. In particular, we seek to build a lightweight, image-guided therapy device that would have the capability to ablate tumors, stop bleeding, and to detect, as well as to comminute, small kidney stones—all in a single device. These medical conditions are also commonplace on Earth. Uncontrolled, occult bleeding due to trauma is a major Earth-based risk. In a multi-center review of 72,151 trauma admissions at a civilian health center, 62% of those requiring emergent operative treatment were in shock on arrival in the operating room, and the actual cause of death was credited to bleeding in 82% of the cases. Indeed, exsanguination is often reported as the major cause of mortality (~50%) in combat casualties. Accordingly, we have been undertaking a major effort, funded by a variety of federal agencies (but principally by the DoD) and in partnership with industry (e.g., Philips Medical and Siemens Ultrasound) to develop a portable device that could be used in forward echelons of the battlefield to address the risk of death by exsanguination. We have leveraged this extensive DoD-funded effort in our research for the NSBRI. Furthermore, several companies, in addition to Philips and Siemens have product development programs underway on this concept, viz., THERUS and ACOUSTX, both based in Seattle, and both of which have licensed technology from the University of Washington (UW).
Research Impact/Earth Benefits:	This image-guided therapy approach utilizes the capability of HIFU to rapidly elevate the temperature to coagulative necrosis levels (> 55 oC). The capability to induce coagulative necrosis, particularly under image-guidance, provides a unique opportunity to treat benign and malignant tumors. Two Seattle-based start-up companies, MIRABILIS and UST, have licensed UW technology to develop products that would use image-guided HIFU to ablate tumors. Our NSBRI Research Team has also assisted YUANDE, a Chinese company, in its application to the FDA for an IDE (Investigative Device Exemption) to treat patients with advanced stage pancreatic cancer. Our laboratory has had considerable experience in the treatment of stone disease; we are now participating in the 15th year of an NIH Program Project Grant to study the physical basis of stone comminution and tissue damage in shock-wave lithotripsy. Between 5% and 15% of the general population will develop a urinary calculus during their lifetime, which will account for about 7 to 10 of every 1000-hospital admissions in the United States. One of our major contributions to lithotripsy research has been the determination that acoustic cavitation plays a major role in stone comminution. Our NSBRI project builds upon this extensive experience to develop a system for detecting and comminuting a kidney stone in the unique and remote environment of space. Because fluoroscopy will (probably) not be available in extended space travel missions, our ultrasound-based detection and comminution approach has been developed. In performing this research, we have developed unique techniques and methodologies for monitoring the progress of lithotripsy treatment that appear to have immediate Earth-based applications.
Task Progress:	Tasks 1a-d: To develop a probe applicator containing an embedded image probe...To integrate the probe applicator with a human interface and power unit... To perform a series of experiments... to induce acoustic hemostasis transcutaneously. Two integrated applicators have been developed and tested. The devices are described in recent manuscripts and patent applications. Our NSBRI Fellow published several papers on accelerating hemostasis through the use of ultrasound contrast agents. 2a-c. To develop a real-time guidance control and targeting system for HIFU ablation... To develop a lesion-imaging technique to permit real-time monitoring of the HIFU lesion... of interest in ex vivo and phantom tissues. These 3 tasks have been completed with our image-guided HIFU therapy system. The system relies on the appearance of an echogenic region that indicate the HIFU focus on ultrasound scans. This image-guidance technique is used in the clinical system we are testing; our patent application describes a method and a device to permit real-time imaging while using HIFU. Our recent paper describes a targeting approach that uses a nonlinear acoustics approach to rapidly produce boiling, which in turn produces an echogenic spot at the HIFU focus. In a recent paper, we described tests of an automated system to automatically recognize a bleed on the ultrasound image and target and treat the bleed with a phased array source, stopping when bleeding is no longer detected in the image. 2d: To develop a therapy protocol based upon a validated numerical model... A model, based on linear acoustic propagation, was developed in FEMlab for use in temperature monitoring of HIFU ablation. A more refined model incorporating nonlinear acoustic propagation and using a finite difference method was also tested in in vitro and in vivo experiments. A number of publications, describing these techniques, have been published. 3a-c. To determine the acoustic protocols required to produce cavitation levels sufficient to comminute artificial kidney stones... To measure the acoustic pressure amplitudes and intensities that are produced... To determine the minimal sizes of renal calculi that can be detected.... Dr. Bailey spent approximately 1 month in Japan working with our collaborators on using their two-frequency system to comminute stones; a Postdoc from our Japanese collaborators' lab joined us for 4 months. In 2006 we reported optimized stone comminution within a calibrated exposure range. 3d. To demonstrate the capability of the integrated system.... The fully integrated system for stone comminution has been tested. The stone is localized by twinkling artifact on an ultrasound imager and then treated with the phased array used in the automated system. Our clinical cancer system also comminuted stones in vitro. A patent application was filed for a method to monitor stone comminution.
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